Exp No:13 Experiment to handle missing values and inappropriate data in a given data set.

AIM:

To handle missing values and inappropriate data in a given data set.

Description:

Understand the procedure to handle missing and inappropriate data.

CODE:

```
import numpy as np
import pandas as pd
df = pd.read_csv("student_missing_inappropriate.csv")
df
df.duplicated()
df.info()
df.drop_duplicates(inplace=True)
df
len(df)
index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
index
df
df.loc[df["Marks"] < 0, "Marks"] = np.nan
df
df.loc[df["Attendance"] < 0, "Attendance"] = np.nan
df
df.loc[df["Fees"] < 0, "Fees"] = np.nan
df
df["Department"].unique()
```

```
df["Marks"].fillna(
   round(df["Marks"].median()),
   inplace=True
)
df
df["Attendance"].fillna(
   round(df["Attendance"].median()),
   inplace=True
)
df
df["Fees"].fillna(
   round(df["Fees"].mean()),
   inplace=True
)
Df
```

OUTPUT:

	0
0	False
1	False
2	False
3	False
4	False
5	True
6	False
7	False
8	False

	0
9	False

dtype: bool

	StudentID	Name	Age	Marks	Department	Attendance	Fees
0	1	Arun	19	85	CSE	92	50000
1	2	Priya	20	92	CSE	88	52000
2	3	Ravi	19	76	ECE	75	48000
3	4	Divya	20	-5	EEE	91	51000
4	5	Kiran	21	88	CSE	86	53000
6	7	Meena	20	81	ECE	80	-1000
7	8	Ajay	19	95	EEE	95	50000
8	9	Sneha	20	-10	CSE	89	54000
9	10	Vijay	22	90	CSE	-1	55000

	StudentID	Name	Age	Marks	Department	Attendance	Fees
0	1	Arun	19	85.0	CSE	92.0	50000.0
1	2	Priya	20	92.0	CSE	88.0	52000.0
2	3	Ravi	19	76.0	ECE	75.0	48000.0
3	4	Divya	20	NaN	EEE	91.0	51000.0
4	5	Kiran	21	88.0	CSE	86.0	53000.0
5	7	Meena	20	81.0	ECE	80.0	NaN
6	8	Ajay	19	95.0	EEE	95.0	50000.0

	StudentID	Name	Age	Marks	Department	Attendance	Fees
7	9	Sneha	20	NaN	CSE	89.0	54000.0
8	10	Vijay	22	90.0	CSE	NaN	55000.0

	StudentID	Name	Age	Marks	Department	Attendance	Fees
0	1	Arun	19	85.0	CSE	92.0	50000.0
1	2	Priya	20	92.0	CSE	88.0	52000.0
2	3	Ravi	19	76.0	ECE	75.0	48000.0
3	4	Divya	20	88.0	EEE	91.0	51000.0
4	5	Kiran	21	88.0	CSE	86.0	53000.0
5	7	Meena	20	81.0	ECE	80.0	51625.0
6	8	Ajay	19	95.0	EEE	95.0	50000.0
7	9	Sneha	20	88.0	CSE	89.0	54000.0
8	10	Vijay	22	90.0	CSE	88.0	55000.0

RESULT:

The duplicate and inappropriate values are successfully identified and handled and the dataset was cleaned using pandas